

Updating Cleverscope CS328A, CS448 or CS548

1 Scope

This document shows how to update the **Cleverscope** application using the latest version from the **Downloads** section on our website and how to update the firmware inside a *Cleverscope Acquisition Unit*.

2 Important – read before starting

The **ROM Loader** has changed. From now on please use **ONLY** Rom Loader x64.exe, Version 8.61 or later, as included with this update.

This **ROM Loader** uses files, with a name format based on the type of scope. These are:

- CS328 or CS328A cscope_fw 6573 6473 5338.cmf
- CS448 v1.0 cscope_fw 7030.cmf or later
- CS448 v1.2 cscope_fw 7230.cmf or later
- CS448 v1.3 cscope_fw 7330.cmf or later
- CS448 v1.4 cscope_fw 7430.cmf or later
- CS548 v1.5 cscope_fw 7549.cmf or later
- CS548 v1.6 cscope_fw 7650.cmf or later

The Rom Loader application automatically works out which version to update, and will only display files that match the version.

Update to Calibration Table

If you are updating a very old CS328A (version 6441 or earlier) you will need to do a *Once Yearly* calibration followed by *Standard* calibration. Instructions are in section 5, after updating the firmware.

Automatic Full or Update Install

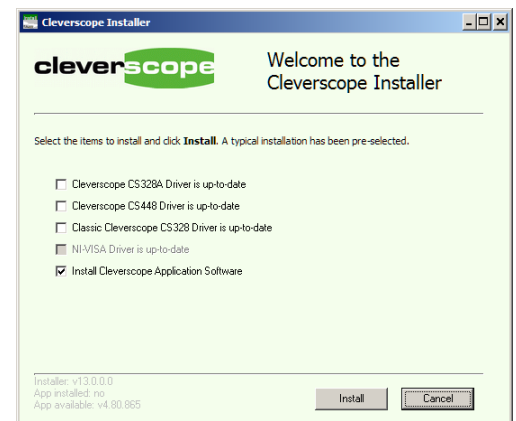
Our installer installs either a full install (if there is no install on the PC) or an update (if there is already an Install). Please do **NOT** un-install first.

3 To update the Cleverscope Application

Download the latest Installer from the **Downloads** section on our website, and extract the folders in the zip to your hard drive. Do not install out of the zip. See section 6 for more help on this process.

Please turn **off** the *Cleverscope Acquisition Unit* before doing the application update. This ensures the correct drivers are loaded.

Run **setup.exe** out of the folder. Accept the defaults. Click *Install*. The update should happen automatically. Please accept any security warnings.



If the website update description says you need the latest firmware, install the firmware as shown in section 4.

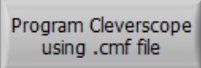
4 To update the Cleverscope Acquisition Unit firmware

Connect and power up your Cleverscope Acquisition Unit (CAU). You can use either an Ethernet or USB connection, for both the CS328A, CS448 or CS548.

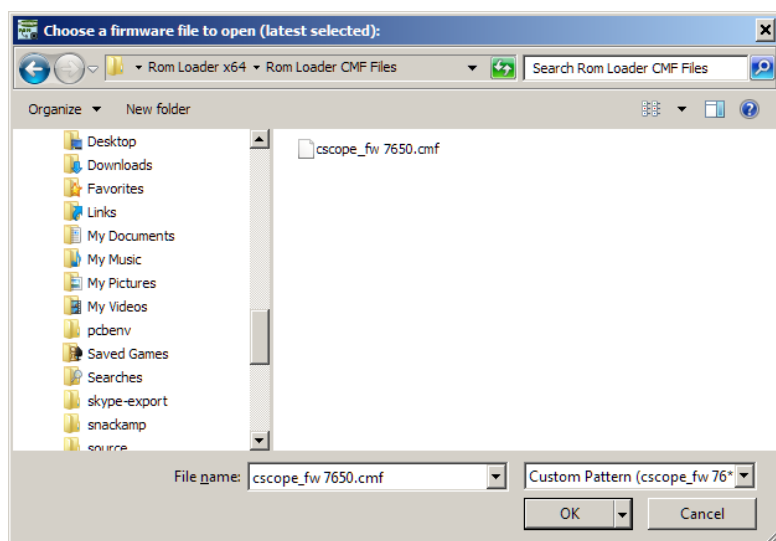
Please use Cscope ROM Loader 8.61 or later. You will find this in the Start/Cleverscope4 x64/Rom Loader x64 menu. Click **Cscope Rom Loader.exe** to start.

If you are using Ethernet, and there are other Cleverscope's on your network, use the Choose USB/Ethernet button to choose the correct Cleverscope. If you have only one unit, you don't need to do anything.

Click the button.



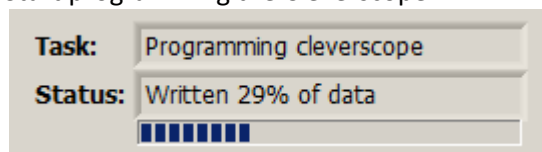
This will bring up a file dialog for you to locate the correct .cmf file. The default is your My Documents\Cleverscope\, and the file will be automatically selected:



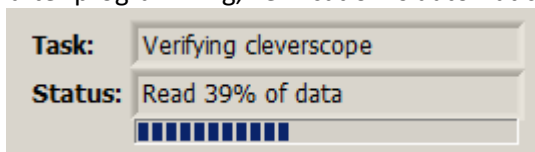
Only the files that match your Cleverscope will be displayed.

Select the file and click OK. It may take a moment or two, and then Rom Loader will

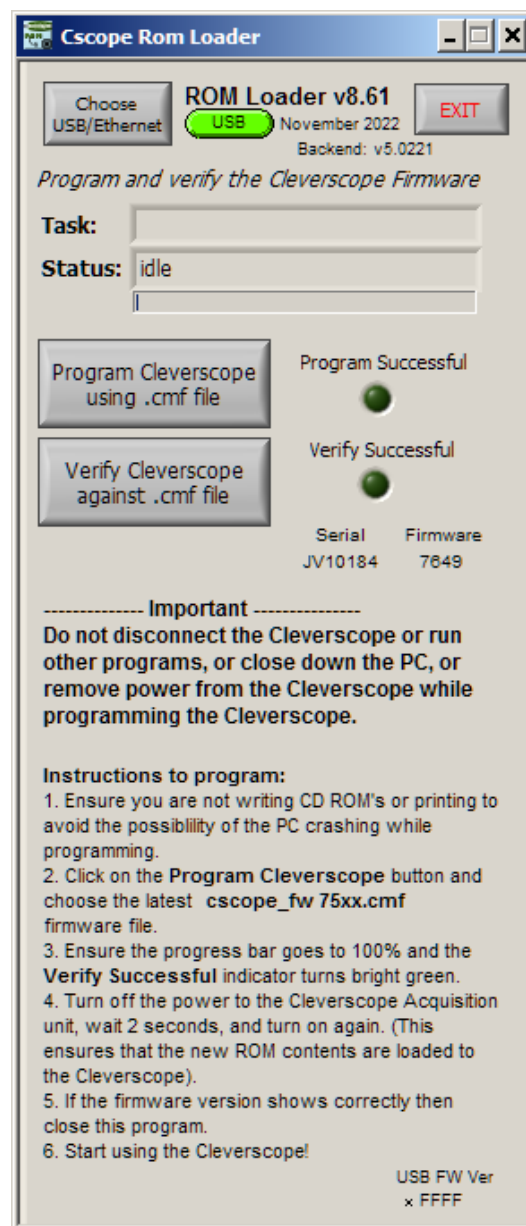
Start programming the Cleverscope:

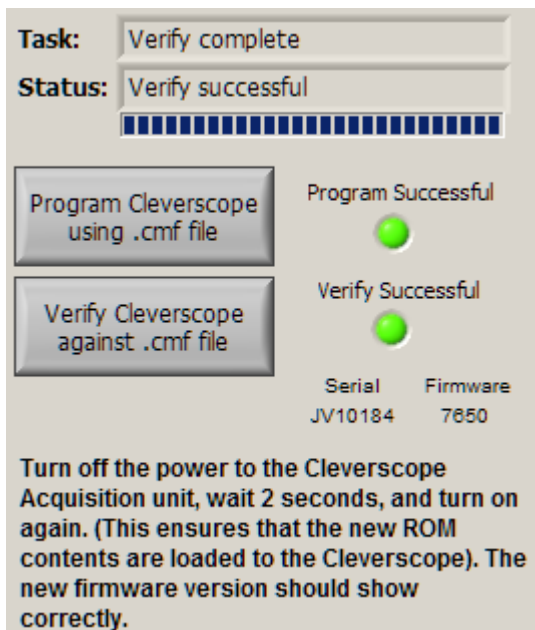


after programming, verification is automatically started:

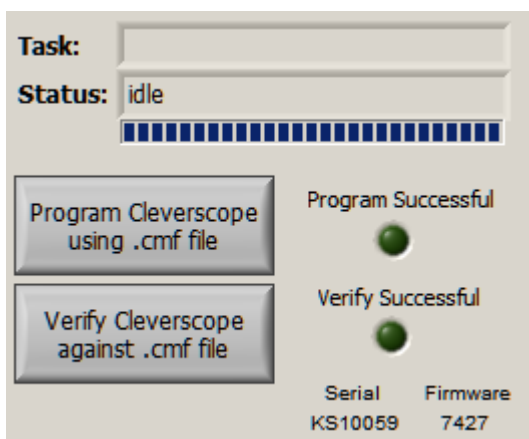


After verification, you will see a message to turn the Cleverscope off, and then on again:





After you do that, the Rom Loader should automatically update the Firmware version to the one you just programmed.



Well done! You have now updated both the application, and the firmware.

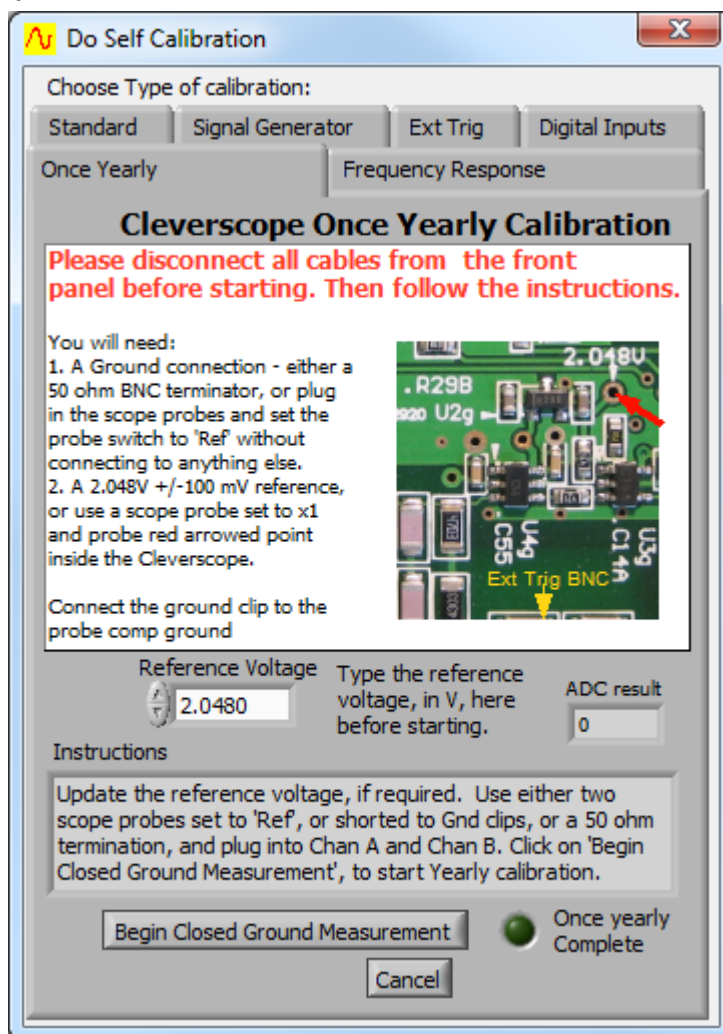
If you CS328A firmware was version 4642 or earlier, we strongly suggest that you do a Yearly Calibrate, as shown in section 5.

5 Calibrating your Cleverscope CS328A

If you are upgrading from a version prior to 6442 to 6442 or later, you will need to do a one-time full calibration. The calibration table has changed, and the calibration procedure will update it.

To carry out the calibration follow these steps:

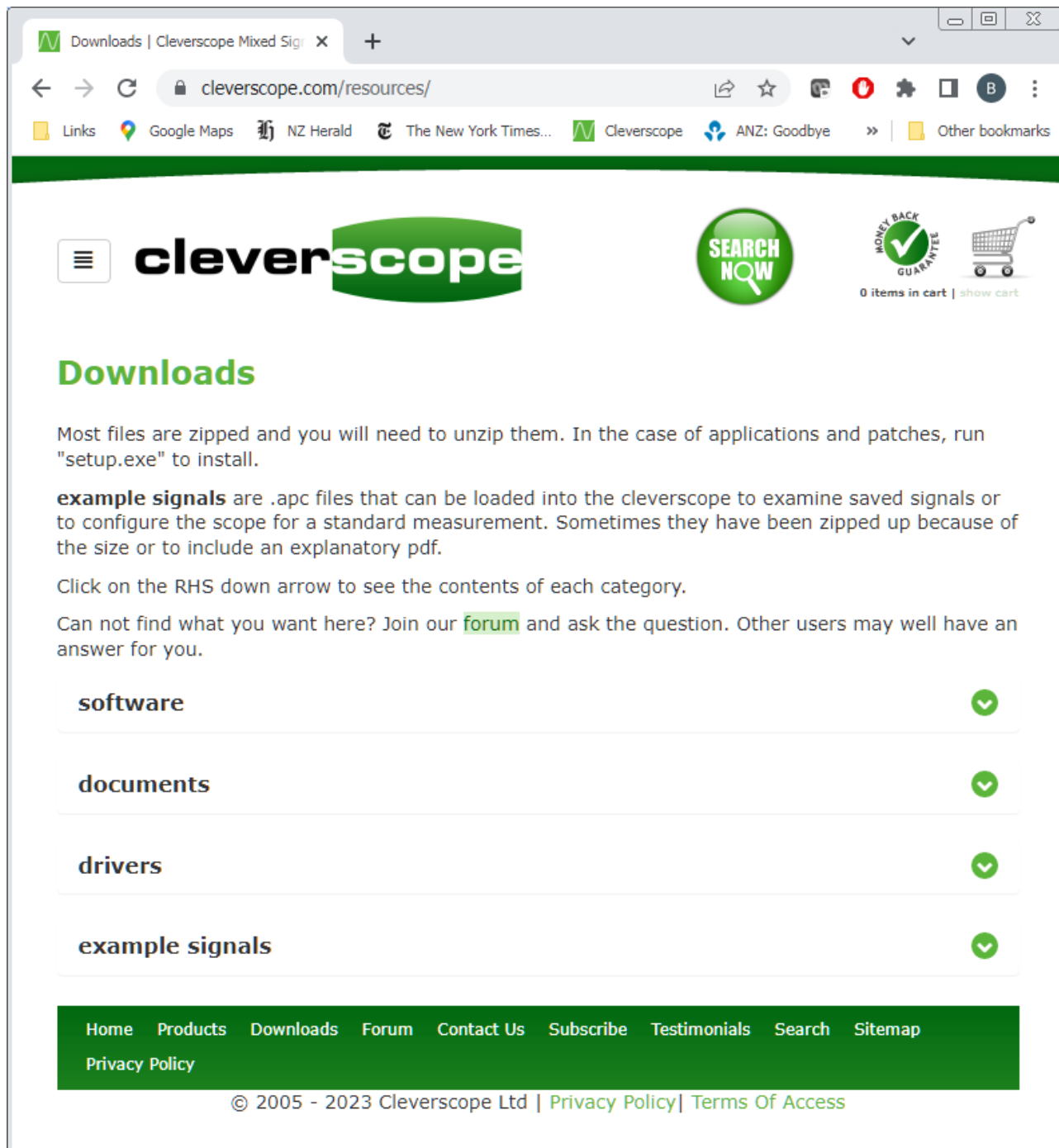
1. Use a Philips screwdriver to undo the two screws under the *Cleverscope Acquisition Unit*. Remove the lid, and power up. Leave it powered for 30 minutes to become temperature stable before calibrating at step 5 below.
2. Update your **Cleverscope** application to the latest.
3. Ensure your **Cleverscope Acquisition Unit** has the latest firmware in it.
4. Run the application, and use **Auto** to acquire. You should see a trace on the screen.
5. Choose the menu **Settings/Calibrate Cleverscope**.
6. Click the **Once Yearly** tab
 - a. Take two Cleverscope scope probes and set the probe switches to *Ref*. This shorts the probe tip to ground. Connect one to Chan A and the other to Chan B. Do not connect the probe tip to anything else. Click **Begin Closed Ground Measurement**.
 - b. When the button changes to **Begin Open Measurement**, remove the probes, and click **Begin Open Measurement**.
 - c. When the button changes to **Begin Ref A Measurement**, set the probe switch to *1x*, plug one probe into Chan A, connect the ground clip to the front panel test pin with the earth symbol above it, pull off the grabber tip if fitted, and probe the via or component pointed at by the red arrow in the picture. Use firm pressure. It is about 40mm in from the front panel Ext Trig BNC. Click **Begin Ref A Measurement**.
 - d. When the button changes to **Begin Ref B Measurement** move the probe to Chan B, and click **Begin Ref B Measurement**.
 - e. When the button changes to **Finish** you have completed the *Once Yearly Calibration*.
7. Now carry out a standard calibration. Click the **Standard** tab. Remove the scope probe from the front panel. Click **Start Self Calibration**. Have a cup of coffee! When completed click **Finish**. Do an **Auto** to ensure correct operation. You may wish to carry out a Sig Gen calibration if you have one installed in your *Cleverscope Acquisition Unit*.
8. Replace the lid, being careful to correctly trap the front/rear labels in the groove by reverse turning 1/2 a turn, and do up the screws. You are done.



6 How to download the Installer from our website

You may download a copy of the latest Cleverscope application software from the **Downloads** page of the Cleverscope website at www.cleverscope.com. The following explanation is for **Windows 7**. The process will vary slightly for other operating systems.

Visit the www.cleverscope.com site, and then go to the **Downloads** page shown below.



Click on the drop down button next to software. The CS302 Multichannel Install x64 is generally the best one to choose. If you are using a CS328A, on an x32 platform, Use CS300 2 Channel Install. Both installers automatically do either a full install (no prior installation on the PC), or an update (if an installation is already on the PC). Do **not** un-install first.

Downloads | Cleverscope Mixed Sigr x

cleverscope.com/resources/

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Downloads

Most files are zipped and you will need to unzip them. In the case of applications and patches, run "setup.exe" to install.

example signals are .apc files that can be loaded into the cleverscope to examine saved signals or to configure the scope for a standard measurement. Sometimes they have been zipped up because of the size or to include an explanatory pdf.

Click on the RHS down arrow to see the contents of each category.

Can not find what you want here? Join our [forum](#) and ask the question. Other users may well have an answer for you.

software

CS302 Multichannel Install x64

10 Mar 2023 - 429 MB

This is the recommended software for all our scopes. We have moved to x64 bit. **Note there is an issue with AMD Processors that requires a new environment variable - see [AMD](#)** Version 5.2.21 Cleverscope 4 x64 controls the CS448, CS328A, CS320A and classic CS328. It supports calibration of the CS328A and CS448 scopes (not the Classic CS320). Includes ROM updates for all version of the CS448 and CS328/A. Adds options for the CS1133 Vsat probe selection..

This is a full install or update. Unzip directory to your hard drive. For updates DO NOT uninstall the software first. Run setup.exe.

[Download](#)

CS300 2 Channel Install

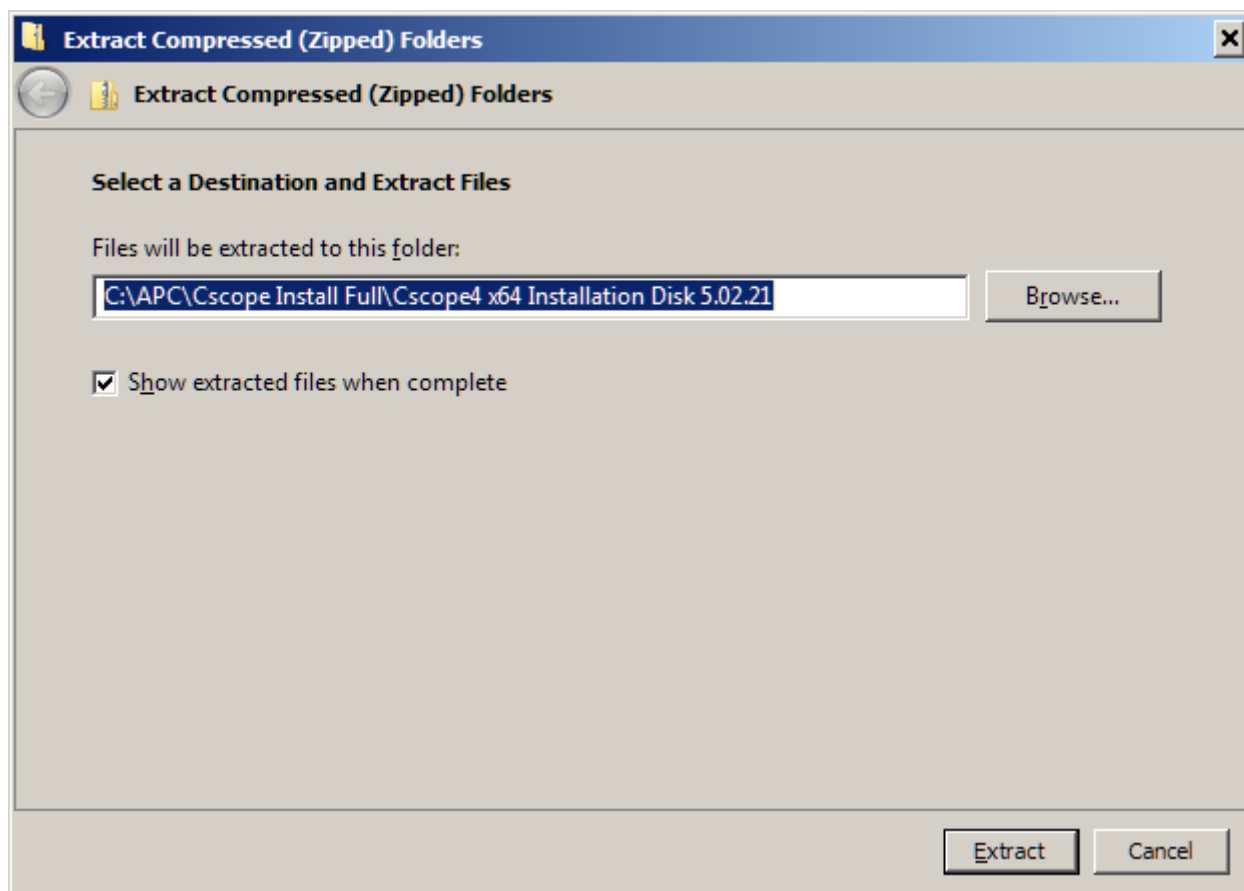
19 Jan 2023 - 242 MB

Full v4.69995 Installer for Win XP, Vista, Win 7, Win 8 and Win 10 x32 and x64. Works with all variants of Cleverscope two channel acquisition units. Does NOT work with CS448 or CS548. Use for an original install or an update. DO NOT uninstall the existing installation first. Use this application to calibrate a two channel scope. This version supports DDE transfer of FRA results. Does triggered FRA start. Extract folder in zip to your hard drive and run setup.exe. Make sure your firmware is 6473 or later.

[Download](#)

Click download and allow the file to be downloaded as normal. Make a directory for the install, and copy the contents of the zip file (which is in the form **Cscope Installation Disk 4808xx.zip**) to that directory.

Navigate to the file (usually in your Downloads folder), and right-click on the zip file and choose **Extract All**:



Click Browse, and locate the folder you created, and click **Extract**.

Note The zip file **must** be extracted because the installer (*Setup.exe*) will not run from inside the zip.

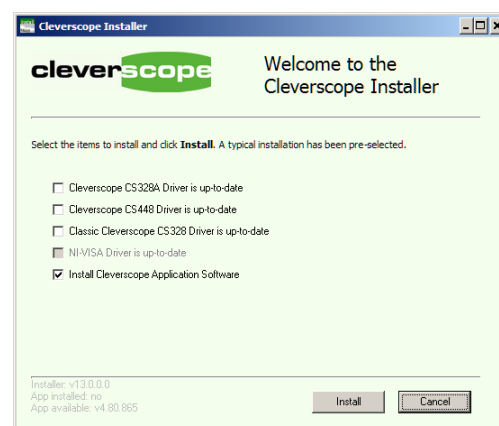
Open the *Install* folder then run **Setup.exe**. The **Cleverscope Installer** window will open, and four or five choices are presented. Those choices that are already up-to-date will not be checked. However, you can still force an update if required by checking the option. If you are updating drivers, ensure the *Cleverscope Acquisition Unit* is turned **off**.

Note Ensure **Install Classic CS328 Driver** is checked if you have the *Classic CS328* (with a serial number less than 4000).

Ensure **Install Cleverscope Application Software** is checked and click **Install**. The installer will update the current copy of **Cleverscope** and any drivers selected. If required, please click **Continue Anyway** while installing the USB drivers. Trust any drivers. Once finished, click **Finished** to exit the installer.

This completes the application update process. This process does **not** update the firmware. See section 4 for updating firmware.

To use the *Cleverscope Acquisition Unit*, plug it in, and turn on the power. If the drivers were updated, the driver will be automatically loaded in **Windows 7 or Windows 10**.



6.1 Using the Cleverscope application with a Classic CS320/8

If you are updating the **Cleverscope** application, and you plan to use the application with a *Classic CS320/8*, and you have not used the *Classic CS320/8* before, you will need to install the **Classic Cleverscope CS328 Driver** by checking the option. *Classic Cleverscopes* have a serial number of less than 4000.

6.2 Updating Cleverscope applications prior to 6430.

If your **Cleverscope** application had a version number prior to 6430, it may have been using the legacy *QuickUSB* driver (CS328A varieties only). This driver is no longer used. If the driver is not required by another application, an additional option will be displayed **Remove Legacy QuickUSB Driver**. If this option is selected, the *QuickUSB* driver will be removed. Make sure you **restart** the computer after the update is complete.

7 Install Cleverscope for the first time

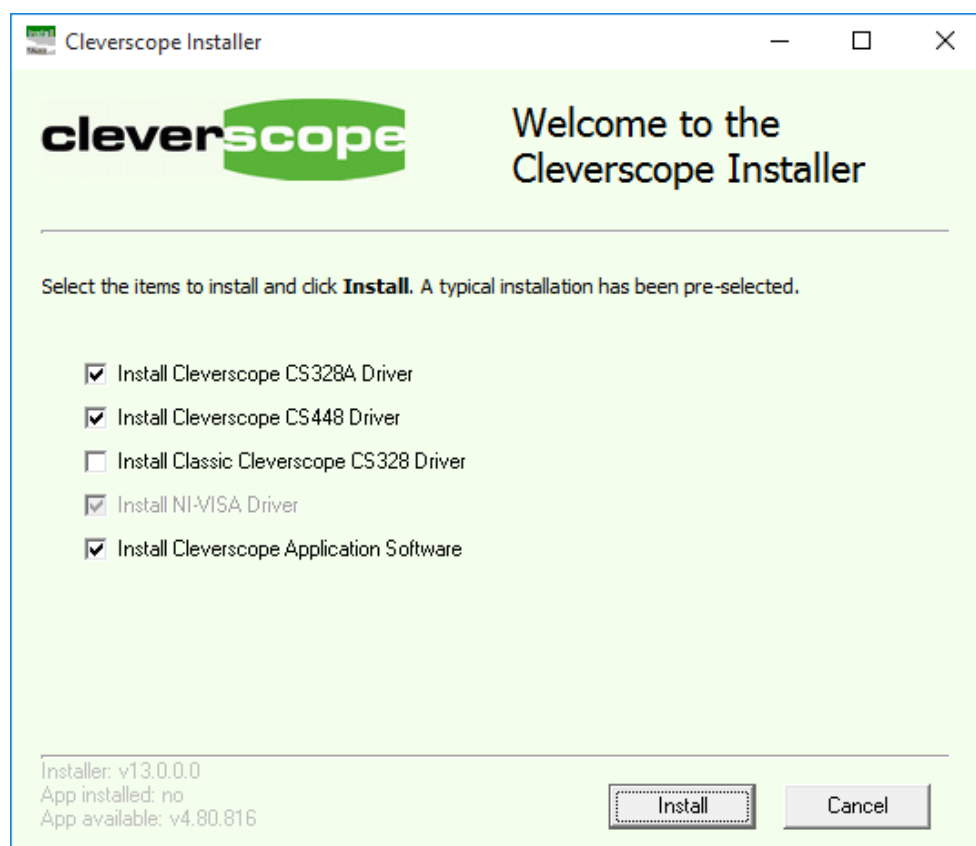
Download the Install as shown in section 6.

See <https://cleverscope.com/warranty/> to extend your warranty period.

Installation Steps

These are the steps to install the CS548 or CS328A on your system. You can install the CS548 on Windows 7.1, 8, 8.1 and 10. The CS548 is not supported under Windows XP or Vista. The CS328A is supported for all operating systems from Windows XP on. App version 480851+ supports both CS548 and CS328A, but it does not calibrate the CS328A. For that you will need to install App version 469993, which only supports CS328A.

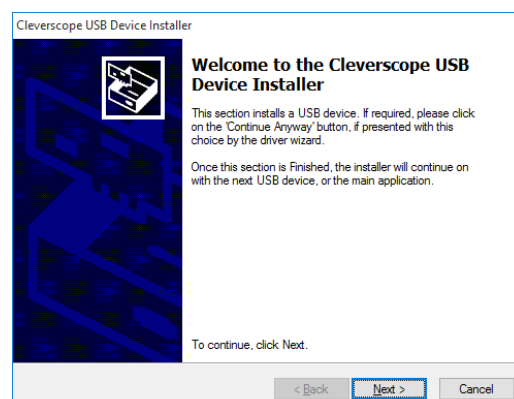
1. Download the install, and extract as described in Section 6. Run **setup.exe**. You should see this:



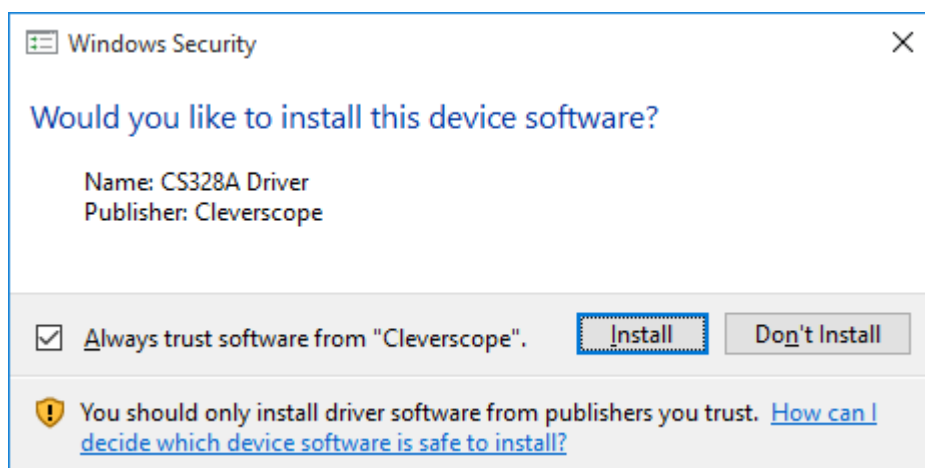
If you are planning on using a CS328A as well, select 'Install Cleverscope CS328A Driver', otherwise deselect this option. Click **Install**.

2. This dialog will appear for each driver:

Click **Next**

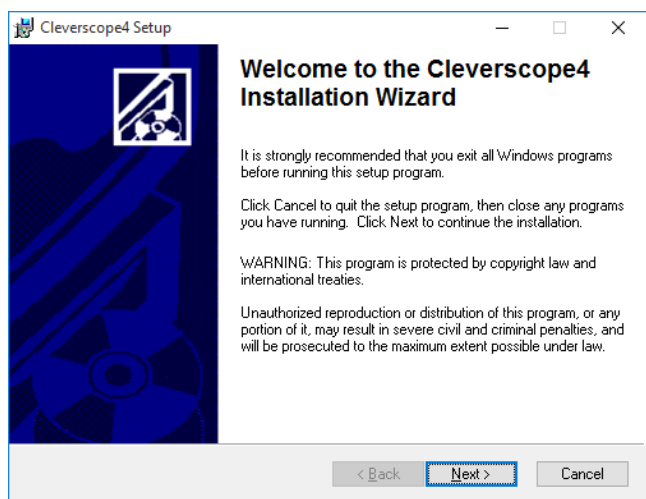


3. You will get a security warning for each driver. Please click 'Always trust software from Cleverscope, and click *Install*:



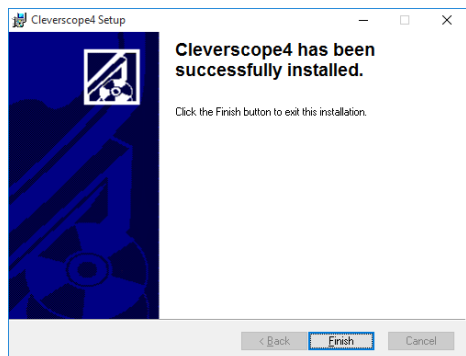
4. Do this for each driver - there may be three: CS328A, CS548 and NI-Visa. Please tick the 'Always trust..' option for each one. Click *Finish* for each driver, as the driver installer finishes.

5. After the drivers are installed, the Cleverscope4 application will install:



Click *Next*, and you are given a choice of installation location. We recommend leaving it as the default.

6. On completion of the installation you should see this dialog:



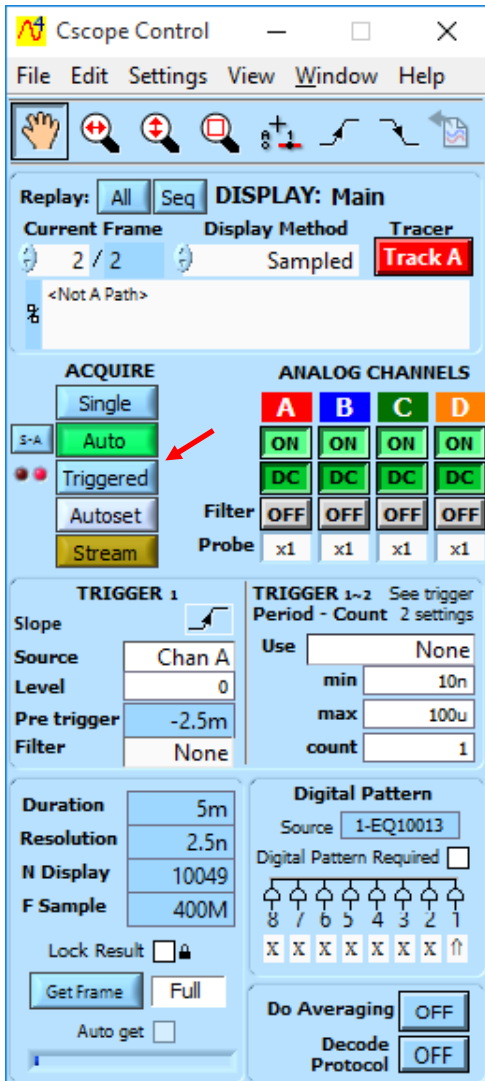
Click *Finish* to complete the installation. Well done!

7. Connect the CS548 to power, without plugging in the USB cable. First the Red LED will come on during the system test. This should go out, and the green LED will flash until the digitizer and I/O system are activated and calibrated. When the green LED stops flashing, and is on solid green, the system is ready.

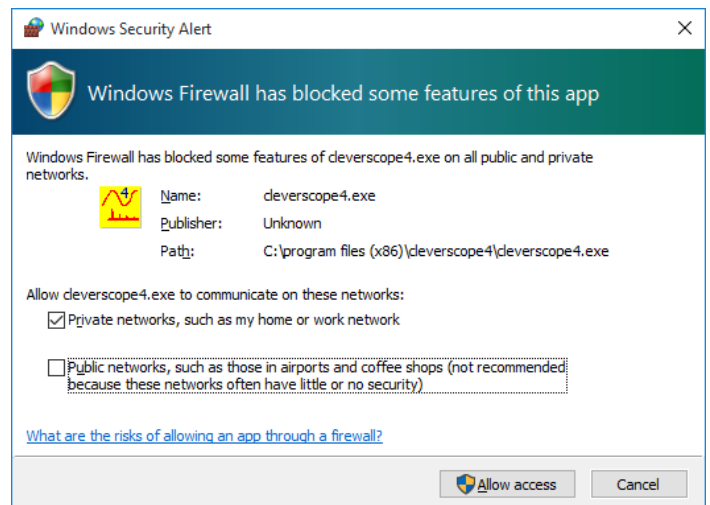
8. Plug the USB 3.0 cable into the PC and the CS548 (or the USB 2.0 cable for the CS328A). This will activate the driver. For older Win 7 versions you may be asked to restart the operating system. Do this if required.

9. Choose Windows Start/All Programs/Cleverscope4/Cleverscope4. You may also see Cleverscope4 in the quick start list. For Windows 8 and 8.1 you should find Cleverscope4 in the recently installed list, otherwise you will need to search for it.

After starting Cleverscope4, click on the *Auto* button (arrowed, which highlights in green):



The system may give you a warning that it requires Ethernet access. The Cleverscope app can communicate with other Cleverscopes over a network.



If you wish to control a Cleverscope over a network (for example Ethernet connected Cleverscopes), click 'Allow Access'. The USB version does not need access.

Now Open the manual using Start/Cleverscope4 and choose the manual. It is still the CS328A manual, and it will be updated soon, but operation is similar. Follow section 3. Getting started.

We have guided video tours of the existing CS328A app here: <https://cleverscope.com/tours/>

Choose the 'Start Using Cleverscope' tour.

7.1 Update a first generation installation

If your application version number begins with 3.2 or 3.3, you have an old run time library. You will need to do a new install to obtain the new run time library.

Please carry out the following steps:

1. Rename the 'Cleverscope' program files folder to 'Cleverscope_old'. Do **not** un-install the old application.
2. Carry out the steps in **7 Install Cleverscope for the first time** above.
3. Update the firmware in your CAU by following the steps in section 4.

If you are one of the very few customers with a CAU with a serial number beginning with 'A00' please contact Cleverscope at support@cleverscope.com before attempting an upgrade.

7.2 Using Windows 98SE, ME and NT

If you use Windows 98SE, ME or NT you will need to download the correct installation or update from the Cleverscope website. You can find these at www.cleverscope.com/resources/. These operating systems are no longer supported, and are only available for the CS328 *Classic*.